The Effects of Global Uncertainty Shocks on the Brazilian Economy

João P. M. Silva

Universidade Federal de Pernambuco

April 9, 2025

Content

Introduction

Introduction

- **What is uncertainty?**

- Knight (1921) coined the modern definition of uncertainty and argued that uncertainty is common in economic decision making.

- He differentiates between two concepts that influence the economic behavior of individuals and firms regarding future outcomes:
    1. **Risk:** refers to situations where a known probability distribution governs a set of events.
    2. **Uncertainty:** refers to situations where individuals cannot assign probabilities to the possible outcomes or one does not know all the possible outcomes, thereby limiting their ability to make fully informed decisions.

Introduction

- **How does economic uncertainty affect the economy?**
- A growing body of literature seeks to answer this question, particularly following the seminal work of Bloom (2009).
- The surge in uncertainty following the 2008 global financial crisis—and its potential role in slowing the recovery process—has drawn increased policy attention to the issue. (Bloom, 2014)
- It is common to attribute negative economic results, represented by the rise in inflation, slower growth and rise in uncertainty, to domestic or foreign economic policies.
- Although arguments agree about the negative relation between uncertainty and economic activity, this association still lacks empiric evidence, especially in a developing country such as Brazil. (Costa Filho, 2014)

Introduction

- This present work aims to present empirical evidence on how global uncertainty affects the Brazilian economy.
- We use the novel World uncertainty Index developed by Ahir et al. (2022) and employ a SVAR approach similar to Barboza and Zilberman (2018) based on Baker et al. (2016).
- We aim examine how a global uncertainty shock influences domestic uncertainty and its transmission channels to Brazilian macroeconomic variables.

Preview of Results

- Our findings align with both international and domestic literature on the subject.
- Specifically, we find that a one-standard-deviation shock to global uncertainty, as measured by the WUI, has countercyclical effects on key Brazilian macroeconomic indicators.
- Key results are:
  1. Real GDP declines by 0.004% for at least three years following the shock;
  2. Inflation initially rises 0.07% immediately after the shock and declines by almost 0.1% two quarters following the shock but subsequently rises;
  3. exchange rate rises by 0.015% over the next three years following the shock;
  4. domestic uncertainty increases by 0.12% in response to the shock.

Related Literature

Related Literature

There is a growing international literature on economic uncertainty:

- **Defining risk and uncertainty:** Knight (1921), Epstein and Wang (1994), Davidson (1999), Fernandez-Villaverde and Guerron (2020), Bloom (2014), Cascaldi-Garcia et al. (2023).
- **How do we measure uncertainty?**
  - Proxies of uncertainty: Bloom (2009), Bloom (2014), Baker et al. (2016), Fernandez-Villaverde and Guerron (2020), Ahir et al. (2022).
  - Counting occurrences: Alexopoulos and Cohen (2009), Baker et al. (2016), Ferreira et al. (2019), Ahir et al. (2022).
- **Does uncertainty matter for economic activity?:** Bloom (2009), Haddow et al. (2013), Jurado et al. (2015).

Related Literature

- **Uncertainty short-run transmission channels:** Bloom (2014)
  1. Real-options effects: Bernanke (1983), Brennan and Schwartz (1985), McDonald and Siegel (1986), Pindyck (1991), Dixit and Pindyck (1994).
  2. Risk-premium effects: Arellano et al. (2012), Christiano et al. (2014), Gilchrist et al. (2014).
  3. Precautionary-savings effects: Skinner (1988), Romer (1990), Bansal and Yaron (2004).

Related Literature

Literature on the effects of uncertainty for Brazil is still relatively small.

- Costa Filho (2014) and Barboza and Zilberman (2018) aim to understand the effects of economic uncertainty on Brazilian economic activity.
  - Both use the VAR methodology constructed by Bloom (2009).
- Both reach essentially the same conclusion: uncertainty has a **significant contractionary impact** on Brazilian economic activity.
- Ferreira et al. (2019) build upon Alexopoulos and Cohen (2009) and Baker et al. (2016) and construct the Brazilian Economic Uncertainty indicator (IIE-Br).
  - The also conduct a study using a BVAR approach they conclude that uncertainty shocks cause economic downturn in subsequent periods in the Brazilian economy.

Related Literature

- Silva et al. (2022) additionally develop an index for Brazil that can measure the level of economic policy uncertainty in Brazil.
  - Then, using a VAR model with sign restrictions, they examine the impact of uncertainty shocks on a set of Brazilian macroeconomic variables.
  - They also confirm that rising levels of uncertainty have contractionary effects, which lower consumption and have a negative impact on economic activity.

- Finally, Gea et al. (2021) and Melo and Barros (2024)) evaluate economic policy uncertainty in the context of stock market returns in Brazil.
  - Both come to similar results in respect to the negative impacts uncertainty shocks have on stock market returns in Brazil.

Data

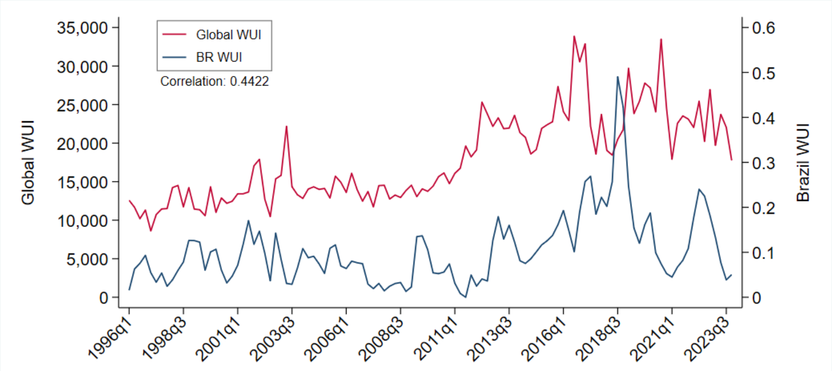
Data and World Uncertainty Index

- Ahir et al. (2022) build the World Uncertainty Index for an unbalanced panel of 143 countries on a quarterly basis starting in 1952.
- The index captures the frequencies of the word "uncertainty" and variants in the Economic Intelligence Unit (EIU) country reports.
- To enable comparison across countries, they scale the raw counts by the total word count in each report, or the amount of "uncertainty" words per thousand words.
- The EIU reports are used for their standardized process and structure, allowing for comparability across time and countries.
- Additionally, the process used to produce EIU country reports reduces concerns about the accuracy, ideological bias, and consistency of the WUI.

WUI Index

Data and world Uncertainty Index

Figure: **Quarterly WUI Indexes for Brazil and Global (1996-2023)**



Note: This figure shows the World Uncertainty Indexes developed by Ahir et al. (2022). The indexes correspond to the Global WUI index and the WUI index for Brazil. The data corresponds to the period of 1996q1 to 2023q4.
Source: policyuncertainty.com

Data and world Uncertainty Index

- Ahir et al. (2022) identify four key stylized facts about the WUI, two of which are particularly relevant to highlight here:
  1. Uncertainty is higher in emerging and low-income economies compared to advanced economies;
  2. Uncertainty is countercyclical.

  - For the first, developing countries experience more frequent domestic political shocks, are more vulnerable to natural disasters, and tend to have less diversified economies, making them more susceptible to external shocks. (Bloom, 2014)
  - For the second, Bloom (2014) emphasizes that uncertainty increases at both macro and micro levels during economic downturns, reinforcing its countercyclical nature.

Data and World Uncertainty Index

Table: Variable Descriptions

Variable	Description	Frequency	Treatment	Source
WUI Global	Global World Uncertainty Index weighted by GDP	Quarterly	Log-Difference	WUI Website
WUI BR	World Uncertainty Index for Brazil	Quarterly	Log-Difference	WUI Website
Real GDP	Real GDP for Brazil	Quarterly	Log-Difference	IBGE
IPCA	Brazilian consumer price index	Monthly	Quarterly average	IBGE
Selic	Brazilian Overnight Interest Rate	Monthly	Quarterly average	BCB
Exchange Rate	Exchange Rate (BRL/USD)	Monthly	Log-Difference and quarterly average	BCB
EPU Global	Global Economic Policy Uncertainty Index	Monthly	Log-Difference and quarterly average	EPU Website
EPU BR	Brazilian Economic Policy Uncertainty Index	Monthly	Log-Difference and quarterly average	EPU Website
IIE-Br	FGV Brazilian Uncertainty Indicator	Monthly	Log-Difference and quarterly average	FGV-IBRE

Data and World Uncertainty Index

- An important observation regarding the WUI indexes is their tendency to spike around election periods. (Ahir et al., 2022)
- WUI typically rises in the quarter preceding an election and remains elevated for one to two quarters afterward. This pattern is particularly evident in Brazil.
- The most prominent spikes in the series align with Brazilian election dates. A notable example is the 2018 election, during which uncertainty surged to an all-time high amid a highly polarized contest between left-wing and right-wing candidates.

Model

Structural VAR Model Specification and Identification

- Main model is a structural Vector Autoregressive (VAR) model with recursive identification.
- Widely used in empirical macro since Sims (1980).
- We use the same approach as Barboza and Zilberman (2018) and Baker et al. (2016).
- The SVAR model provides a practical method of estimating the effects of uncertainty shocks.

Structural VAR Model Specification and Identification

The estimated model has the following format:

$$BY_t = C_0 + \sum_{i=1}^{P} C_i Y_{t-i} + u_t, \qquad (1)$$

Where,

- Y_t is the the vector of endogenous variables;
- B is a 6×6 simultaneous effects matrix;
- C_0 is a 6×1 vector of constants;
- C_i matrices are the lagged coefficients matrices with 6×6 dimensions.
- u_t is the structural shocks vector.

Structural VAR Model Specification and Identification

By left multiplying equation (1) by B^{-1}, we obtain

$$Y_t = \beta_0 + \sum_{i=1}^{P} \beta_i Y_{t-i} + \varepsilon_t \tag{2}$$

were,

- β_0 is a vector of constants;
- β_i denotes a matrix of lagged coefficients;
- ε_t is the the reduced form residuals.

Structural VAR Model Specification and Identification

the structural shocks u_t, are recovered from ε_t:

$$u_t = \begin{pmatrix} u_t^{WUI_G} \\ u_t^{WUI_{BR}} \\ u_t^{GDP_R} \\ u_t^{IPCA} \\ u_t^{SELIC} \\ u_t^{ExRate} \end{pmatrix} = \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} & 0 & 0 \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} & 0 \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & a_{66} \end{bmatrix}^{-1} \begin{pmatrix} \varepsilon_t^{WUI_G\ shock} \\ \varepsilon_t^{WUI_{BR}\ shock} \\ \varepsilon_t^{GDP_R\ shock} \\ \varepsilon_t^{IPCA\ shock} \\ \varepsilon_t^{SELIC\ shock} \\ \varepsilon_t^{ExRate\ shock} \end{pmatrix} \tag{3}$$

Structural VAR Model Specification and Identification

- Eq. (3) also represents the contemporaneous causal ordering of the main estimated model.

- Ordering is based on the principle of prioritizing the most exogenous variables first.

- We order uncertainty variables first, global uncertainty is only impacted by a shock to itself and domestic uncertainty to itself and global uncertainty
  - **Hypothesis:** uncertainty influences macroeconomic variables, with global uncertainty potentially impacting domestic uncertainty, which in turn affects domestic macroeconomic conditions.

- Monetary policy responds to the current state of the economy. In simple terms, M.P. responds to inflation expectations. (Leeper et al., 1996; Christiano et al., 1999)

Structural VAR Model Specification and Identification

- We order the exchange rate variable last in the assumption that the exchange rate tends to be more sensible to a variety of possible shocks in the economy. (Eichenbaum and Evans, 1995)
- Number of lags in the model was selected based on standard lag selection criteria (HQIC and SCIC). We estimated a model with one lag.

Structural VAR Model Specification and Identification

The model has a few limitations:

1. Only rarely economic theory implies on a particular contemporaneous casual ordering. (Demiralp and Hoover, 2003)
2. *"Diversas histórias são possíveis, todas plausíveis."* (Barboza and Zilberman, 2018)
3. This method has the drawback that alternative causal orderings might occasionally be represented by equally convincing stories.

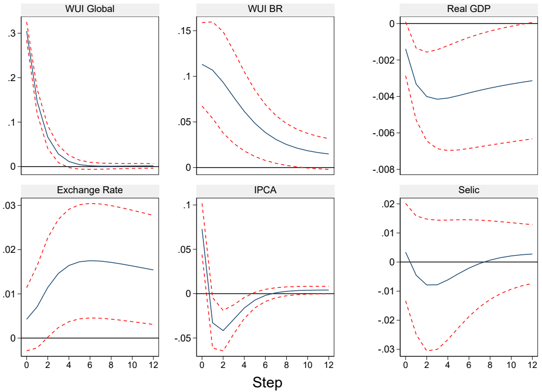
To mitigate arbitrariness in the identification of the model, **many possible ordering or variable alternatives were tested** to check for robustness.

Results

Main Model Results

- Our findings are consistent with both international and domestic literature on the subject.
- We observe that a one-standard-deviation increase in global uncertainty, as measured by the WUI, **has countercyclical effects** on key Brazilian macroeconomic indicators.
- Real GDP contracts by 0.004% for at least three years following the shock.
- Inflation initially rises almost 0.07% immediately after the shock and declines by almost 0.1% two quarters following the shock but subsequently rises.
- Exchange rate increases by 0.015% over the next three years before beginning to decline.
- Domestic uncertainty rises by 0.12% in response to the shock.
- All variables, except the Selic rate, are statistically significant.

Main Model Results



Note: First order SVAR IRF's with impulse from the WUI Global Index with 68% Confidence bands. The IRF's are in response to a one standard deviation impulse.

Main Model Results

- As seen above, global uncertainty shocks have significant impacts on domestic uncertainty.
- Domestic uncertainty is one of the main channels through which global uncertainty affects the domestic economy.
- It's also important to highlight the relatively low response of real GDP to uncertainty shocks.
- Costa Filho (2014) and Barboza and Zilberman (2018) propose that the impact of uncertainty on GDP proxies is notably smaller compared to its effect when estimated using industrial production proxies.
  - Other components of GDP, such as agriculture or services, may not experience the same level of disruption as the industrial sector.
  - Investment is predominantly driven by the industrial sector and is particularly sensitive to uncertainty through the real options effects channel.

Forecast Error Variance Decomposition

- We also conduct a Forecast Error Variance Decomposition (FEVD) analyses.
  FEVD Table
  - The FEVD monitors the volatility as impulses propagate through the system for each period $t \geq 1$.
- Our results show that the contribution of global uncertainty shocks to volatility is relatively low for most domestic variables in the first quarter following the shock.
- However, as time increases, the share of the variance explained by a global uncertainty shock also increases.
- Following 12 quarters after the shock (3 years):
  - global uncertainty shock accounts for 9.1% of domestic uncertainty volatility;
  - 5% for real GDP; 8% for inflation (IPCA); 5.1% for the exchange rate.
  - interest rate (Selic) remains relatively small at 0.21% after twelve quarters.

Robustness

- The results presented in the previous section appear to be robust to various modifications in the model, exhibiting minimal to no changes.
- The following robustness exercises were tested:
  1. Changes in the contemporaneous casual ordering of the main model;
  2. Alternative Model I incorporates the Economic Policy Uncertainty (EPU) indexes for Brazil and the global economy as replacements for the WUI indexes;
  3. In Alternative Model II the Brazilian Uncertainty Indicator (IIE-Br) was incorporated as a replacement for the Brazilian WUI (WUI-BR);
  4. Change in the number of lags to three;
  5. The Alternative Model III excludes the Selic variable.

▸ Robustness Results

Conclusion

Conclusion

- This study aimed to address the following question: How do global uncertainty shocks affect the Brazilian economy?
- To investigate this, we examined the response of key Brazilian macroeconomic variable-including real GDP, inflation, interest rates, and exchange rates—to an uncertainty shock derived from the World Uncertainty Index developed by Ahir et al. (2022) using an SVAR methodology, consistent with Barboza and Zilberman (2018) based on Baker et al. (2016).
- Our findings indicate that uncertainty shocks exert countercyclical effects on Brazilian macroeconomic indicators.
- Our results contribute to the growing body of empirical evidence on the effects of uncertainty in Brazil, a topic that remains relatively underexplored in the literature.

WUI Index

Return

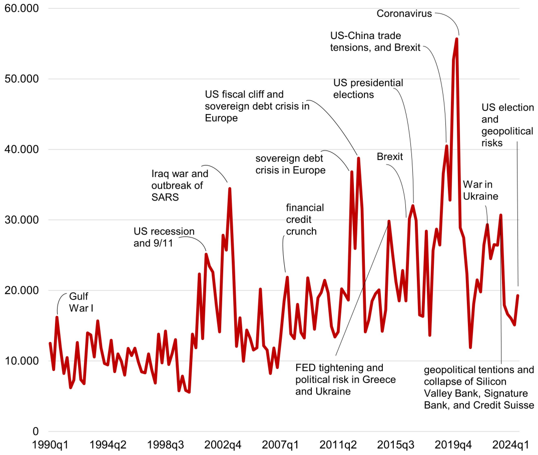


Figure: World Uncertainty Index (WUI) (1960q1 - 2020q1)

Forecast Error Variance Decomposition

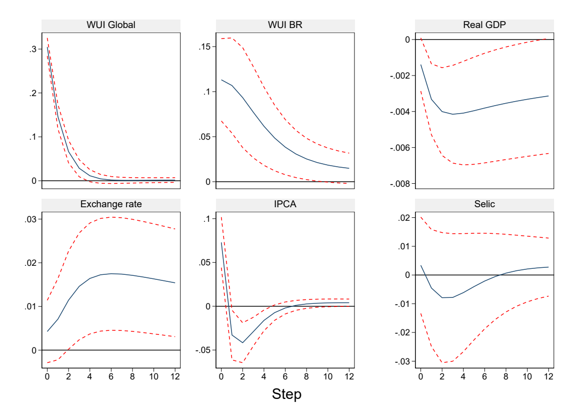
Table: Forecast Error Variance Decomposition

Shock	WUI Global	WUI BR	Real GDP	IPCA	Selic	Exchange Rate
t=1	1	0.054065	0.007976	0.056382	0.000371	0.003225
t=3	0.991169	0.076729	0.039899	0.072235	0.001415	0.012841
t=6	0.978143	0.089152	0.053231	0.080577	0.00232	0.033502
t=9	0.968105	0.091234	0.053412	0.080203	0.002139	0.045348
t=12	0.959342	0.091333	0.05147	0.080137	0.002156	0.05139

Return

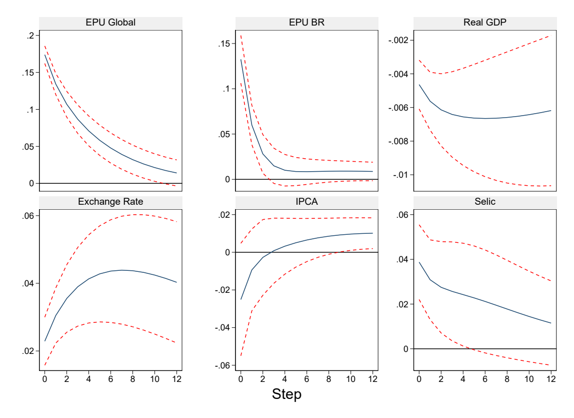
Robustness Results

Figure: *(i)* Second Order IRF's



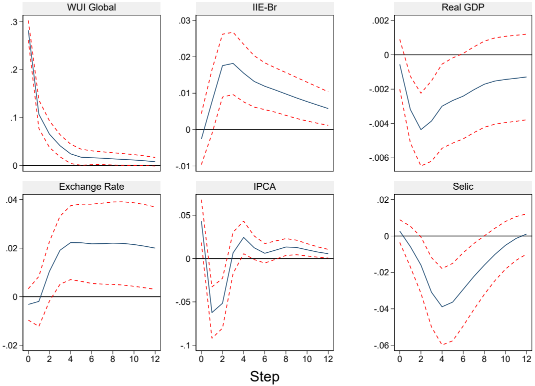
Robustness Results

Figure: *(ii)* Alternative Model I (EPU)



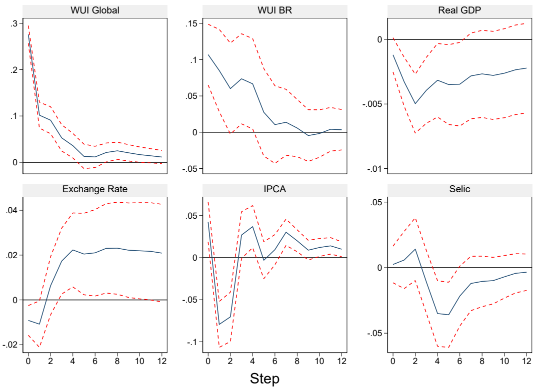
Robustness Results

Figure: *(iii)* Alternative Model II IRF's



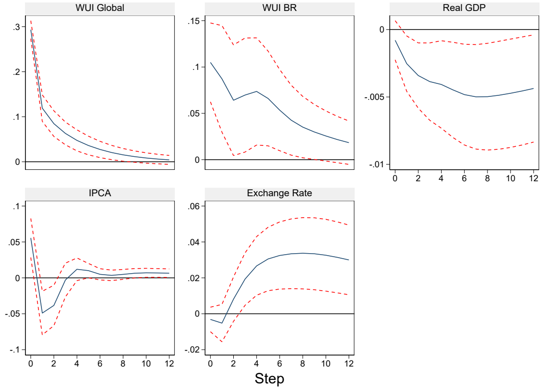
Robustness Results

Figure: *(iv)* Main Model with 3 lags



Robustness Results

Figure: *(v)* Alternative Model III



Return

Ahir, H., Bloom, N., and Furceri, D. (2022). The world uncertainty index. Working Paper 29763, National Bureau of Economic Research.

Alexopoulos, M. and Cohen, J. (2009). Uncertain times, uncertain measures. *University of Toronto, Department of Economics, Working Papers*.

Arellano, C., Bai, Y., and Kehoe, P. (2012). Financial frictions and fluctuations in volatility. *Journal of Political Economy*, 127.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring Economic Policy Uncertainty. *The Quarterly Journal of Economics*, 131(4):1593–1636.

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59:1481–1509.

Barboza, R. and Zilberman, E. (2018). Os efeitos da incerteza sobre a atividade econômica no Brasil. *Revista Brasileira de Economia*, 72.

Bernanke, B. S. (1983). Irreversibility, uncertainty, and cyclical investment*. *The Quarterly Journal of Economics*, 98(1):85–106.

Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica*, 77(3):623–685.

Bloom, N. (2014). Fluctuations in uncertainty. *Journal of Economic Perspectives*, 28(2):153–76.

Brennan, M. J. and Schwartz, E. S. (1985). Evaluating natural resource investments. *The Journal of Business*, 58(2):135–157.

Cascaldi-Garcia, D., Sarisoy, C., Londono, J. M., Sun, B., Datta, D. D., Ferreira, T., Grishchenko, O., Jahan-Parvar, M. R., Loria, F., Ma, S., Rodriguez, M., Zer, I., and Rogers, J. (2023). What is certain about uncertainty? *Journal of Economic Literature*, 61(2):624–54.

Christiano, L., Eichenbaum, M., and Evans, C. (1999). Monetary policy shocks: What have we learned and to what end? In Taylor, J. B. and Woodford, M., editors, *Handbook of Macroeconomics*, volume 1, Part A, chapter 02, pages 65–148. Elsevier, 1 edition.

Christiano, L. J., Motto, R., and Rostagno, M. (2014). Risk shocks. *American Economic Review*, 104(1):27–65.

Costa Filho, A. E. d. (2014). Incerteza e atividade econômica no Brasil. *Economia Aplicada*, 18(3):421–453.

Davidson, P. (1999). *Uncertainty, International Money, Employment and Theory: Volume 3: The Collected Writings of Paul Davidson*. Palgrave Macmillan UK.

Demiralp, S. and Hoover, K. D. (2003). Searching for the causal structure of a vector autoregression. *Oxford Bulletin of Economics and Statistics*, 65(s1):745–767.

Dixit, A. and Pindyck, R. S. (1994). *Investment under uncertainty*. Princeton University Press.

Eichenbaum, M. and Evans, C. L. (1995). Some empirical evidence on the effects of shocks to monetary policy on exchange rates. *The Quarterly Journal of Economics*, 110(4):975–1009.

Epstein, L. G. and Wang, T. (1994). Intertemporal asset pricing under knightian uncertainty. *Econometrica*, 62(2):283–322.

Fernandez-Villaverde, J. and Guerron, P. (2020). Uncertainty shocks and business cycle research. *Review of Economic Dynamics*, 37 Supplement 1:S118–S166.

Ferreira, P. C., Vieira, R. M. B., da Silva, F. B., et al. (2019). Measuring brazilian economic uncertainty. *Journal of Business Cycle Research*, 15:25–40.

Gea, C., Vereda, L., Figueiredo Pinto, A. C., and Klotzle, M. C. (2021). The effects of economic policy uncertainty on stock market returns: Evidence from Brazil. *Brazilian Review of Finance*, 19(3):53–84.

Gilchrist, S., Sim, J. W., and Zakrajšek, E. (2014). Uncertainty, Financial Frictions, and Investment Dynamics. NBER Working Papers 20038, National Bureau of Economic Research, Inc.

Haddow, A., Hare, C., Hooley, J., and Shakir, T. (2013). Macroeconomic uncertainty: What is it, how can we measure it and why does it matter? *Bank of England Quarterly Bulletin*, Q2.

Jurado, K., Ludvigson, S. C., and Ng, S. (2015). Measuring uncertainty. *American Economic Review*, 105(3):1177–1216.

Knight, F. (1921). *Risk, Uncertainty and Profit*. Hart, Schaffner & Marx prize essays. Houghton Mifflin.

Leeper, E., Sims, C., and Zha, T. (1996). What does monetary policy do? *Brookings Papers on Economic Activity*, pages 1–78.

McDonald, R. and Siegel, D. (1986). The value of waiting to invest*. *The Quarterly Journal of Economics*, 101(4):707–727.

Melo, S. B. d. and Barros, L. A. B. d. C. (2024). A new economic policy uncertainty index for Brazil. *Brazilian Review of Finance*.

Pindyck, R. S. (1991). Irreversibility, Uncertainty, and Investment. *Journal of Economic Literature*, 29(3):1110–1148.

Romer, C. D. (1990). The Great Crash and the Onset of the Great Depression. *The Quarterly Journal of Economics*, 105(3):597–624.

Silva, P. H. N., Besarria, C. d. N., and Silva, M. D. d. O. P. d. (2022). Mensurando e avaliando os efeitos de um choque de incerteza da política econômica sobre a economia brasileira. *Economia Aplicada*, 26(3):335–374.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1):1–48.

Skinner, J. (1988). Risky income, life cycle consumption, and precautionary savings. *Journal of Monetary Economics*, 22(2):237–255.